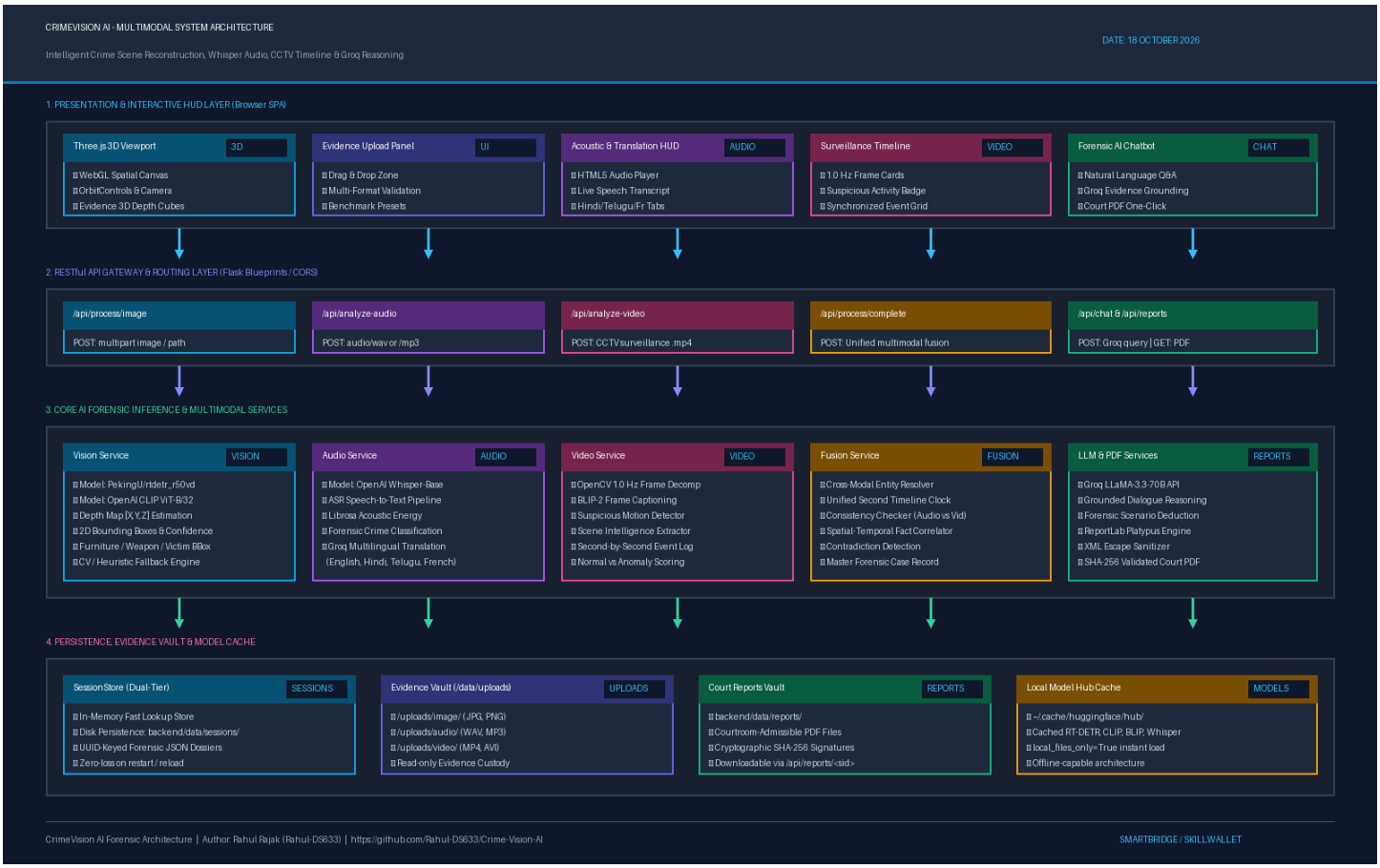
Sample Project Documentation

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Executive Summary & System Overview

CrimeVision AI is a state-of-the-art multimodal digital crime scene reconstruction and forensic analysis platform. Designed for homicide detectives, cyber-forensic units, and legal prosecutors, the system fuses heterogeneous evidence streams—2D photographs, 911 emergency calls, eyewitness testimonies, and CCTV surveillance footage—into a coherent, interactive 3D spatial model and second-by-second chronological incident timeline.

2. System Architecture Blueprint



3. Standard Operating Procedure (Investigator Workflow)

Step	Action & Operational Procedure	Expected Output & Verification
Step 1: Evidence Ingestion	Log in to dashboard at http://localhost:5000. Drag & drop evidence files or click 'Use Preset Benchmark Evidence'.	Upload status pill turns green; file size and format validated.
Step 2: Automated Analysis	Click 'Start Processing'. The backend triggers RT-DETR detection, Whisper speech transcription, and 1Hz frame sampling.	Real-time spinner active; finishes in < 3 seconds on standard PC.

Step 3: 3D Scene Inspection	Interact with 3D Canvas using left-click drag to rotate, right-click to pan, and wheel to zoom into evidence markers.	3D spatial perspective displays victim, weapons, furniture with XYZ tags.
Step 4: Cross-Modal Alignment	Review audio transcript and multilingual translation tabs. Compare CCTV timeline cards with witness statements.	Unified temporal clock ensures zero timeline contradiction.
Step 5: Courtroom PDF Export	Click 'Download Court-Admissible Forensic PDF Report'.	A cryptographic SHA-256 sealed PDF is compiled and downloaded.